

Spot the Error

Two articles that get it nearly right. Nearly is the problem.

Follow-up to *Where the Loops Come From* • MYP • Biology & City Planning

Both articles below were written by someone who read about networks and got excited. Neither is nonsense. Most sentences in them are fine, some are wrong, and several are the more dangerous kind: true-sounding, almost right, and stronger than the evidence allows.

Your job. Find the faulty sentences, label what *kind* of fault each one is, and write a version you could defend.

Fault codes

F	Flat error	The fact is simply wrong.
O	Overclaim	Roughly true, but stated far more strongly than anyone has shown.
C	Missing condition	True only in certain circumstances, which are not stated.
R	Reversed cause	The two things are linked, but the arrow points the other way.
I	Smuggled intent	Something is described as wanting/choosing/helping when nobody has shown it does.
N	Does not follow	The evidence given is real, but the conclusion drawn from it is not supported.

Article A “Nature’s Internet”

Every tree in a forest is connected to every other tree by fungal cables.^[1] These networks were discovered by scientists who proved that mother trees deliberately send sugar down to their own seedlings.^[2] No serious scientist disputes this any more.^[3]

Carbon has been detected moving between trees underground, so we know that trees share food with one another.^[4] The fungi evolved these networks in order to help the forest survive as a whole.^[5]

The same intelligence appears in slime mould. A slime mould has no brain, so it must be using some kind of thinking that science cannot yet explain.^[6] In a famous experiment, a slime mould designed the Tokyo rail network,^[7] recreating the real map exactly.^[8] Because it can also solve mazes, this proves that a single cell can plan ahead.^[9]

Loops are the secret. Loops are always better than branches, because a looped network is stronger.^[10] This is why leaf veins branch outwards like little trees, showing that nature prefers the shortest possible network,^[11] and why the blood vessels in your body are branching trees exactly like rivers.^[12]

No.	Code	What is actually wrong	A sentence you could defend

A1. Two of the twelve sentences are wrong in the *same* way, even though they are about different things. Which two, and what is the shared fault?

Article B “Why Cities Should Copy the Slime Mould”

Cities should be redesigned to copy the slime mould, which always produces the best possible network.^[13] Researchers have shown that doubling a city’s population exactly doubles its economic output^[14] while halving the roads it needs,^[15] and this law holds for every city on Earth.^[16]

The lesson for planners is obvious: stop planning.^[17] Networks organise themselves, so a city left alone will arrive at the right answer.^[18]

Grids are the most efficient shape for getting around, which is why New York was built as a grid.^[19] Medieval street tangles, by contrast, are simply the result of bad planning.^[20] Modern cul-de-sac suburbs are full of loops, just like a slime mould,^[21] and building more roads always reduces congestion,^[22] so a ring road makes a city more efficient in every way.^[23]

Since cities and slime moulds both build loops, and the mathematics is identical, a good city network could be calculated once and then built.^[24] Cities, in the end, are alive.^[25]

No.	Code	What is actually wrong	A sentence you could defend

B1. Sentence 24 contradicts the whole argument of the article you read last lesson. Explain the contradiction in two lines.

B2. Sentence 17 does not follow from sentence 16 even if sentence 16 were true. Say why. Then give one thing a city needs that no amount of self-organisation will supply.

Part C Rewrite it properly

Choose **one** paragraph from either article. Rewrite it so that every sentence is defensible. You may shorten it, and you should: honest writing about this topic is usually shorter than dishonest writing about it.

Rules: keep any fact that is genuinely established, state the condition wherever a claim depends on one, and remove any word that gives something an intention nobody has demonstrated.

Paragraph chosen: _____

C1. How many words was the original paragraph? _____ Your version? _____

C2. What did you lose by being accurate, and was it worth losing?

Part D The harder question

1. Both articles are *more exciting* than the truth. Explain, in your own words, why an overclaim spreads more easily than a careful sentence.

2. One of the two articles could cause real harm if a person in power believed it. Which one, and what harm?

3. Article A treats a fungus as though it were being generous. Suppose instead that the fungus is acting entirely in its own interest, and the trees are its customers. Does anything in the observations change? Does anything in the *explanation* change?

4. Write the single sentence you think both articles were reaching for and failed to say.

Teacher key

Article A

- 1 (O).** Fungal networks connect *some* roots, patchily. “Every tree to every other” has never been shown.
- 2 (I, O).** The famous experiments show movement through shared fungal pathways; deliberate maternal feeding is an interpretation, and a disputed one. Karst, Jones and Hoeksema (2023) found the best-known claims weakly supported and repeatedly amplified by later citation.
- 3 (F).** Flatly false, and the giveaway phrase: appeals to unanimity usually mark the point where the evidence stopped.
- 4 (N).** Carbon moving underground does not tell you the direction, the quantity, the route, or who benefits. It may move through the fungus without ever being shared between trees at all.
- 5 (I).** “In order to” gives the fungus an aim. A fungal network that benefits itself explains the same observations with nothing added.
- 6 (N).** Absence of a brain is not evidence of hidden thinking. The rule Physarum follows is simple, known, and sufficient.
- 7 (F).** It did not design anything. Researchers placed food at town positions and measured the network that grew.
- 8 (F).** Not exact. Comparable in three measured properties: total length, transport efficiency, fault tolerance.
- 9 (O).** Solving a maze by spreading everywhere and withdrawing from dead ends requires no forward planning whatever.
- 10 (C).** Only when demand fluctuates or links can fail. Where demand is fixed, a loop is pure waste, which is the whole point of the river.
- 11 (F).** Leaf veins are heavily looped. The sentence states the opposite of the observation and then draws a conclusion from it.
- 12 (F).** Capillary beds are richly looped, for exactly the reasons the article should have given.
- A1.** Sentences 4 and 9 share the fault: a real observation, and a conclusion that does not follow from it. Accept 5 and 2 as a pair on smuggled intent if defended.

Article B

13 (O). The mould produces a network comparable with a good one on a few measures. Nobody has shown it to be optimal.

14 (F). The reported finding is that output grows *faster* than population, not in step with it. **15 (F).** Infrastructure grows more slowly than population, which is not the same as halving. **16 (O).** Contested. Other analyses question how universal and how predictive the scaling claims are. **17 (N).** Even if cities show regularities without a single planner, it does not follow that planning is useless. Regularity is not the same as a good outcome. **18 (O, N).** “The right answer” assumes there is one and that self-organisation finds it. Self-organised street systems produced cholera as readily as charm. **19 (R).** The arrow is backwards. Manhattan’s grid was laid out largely because rectangular plots are simple to survey, divide and sell; ease of travel was a consequence, not the reason. **20 (C).** They solve a different problem under different constraints: defence, land ownership, terrain, and walking rather than driving. **21 (F).** Cul-de-sac suburbs are deliberately tree-like. That is the definition of a cul-de-sac. **22 (F).** Extra capacity can attract extra traffic, and in some configurations an added road makes journeys longer for everyone. **23 (O).** “In every way” is the tell. A ring road helps some journeys and can gut the places it bypasses. **24 (N).** This is the contradiction worth spending time on: the reason cities need loops is that demand keeps changing, so a network calculated once for today’s demand defeats its own purpose. **25 (N).** A shared mathematical pattern is not shared biology. Rivers and lightning branch alike and neither is alive.

B2. Expected: sanitation, clean water, schools, flood defence, or anything nobody profits from providing individually. Self-organisation optimises what individuals are each trying to get, and is silent on everything else.

Part D notes. Question 2 usually splits the room, which is the point: A is misleading about a forest, B is misleading about decisions that get made with public money. Question 3 is the deepest one on the sheet, since the honest answer is that the *observations* do not change at all and only the story around them does, which is what makes the story so hard to test.

Marking stance. Reward a student who defends a sentence you had listed as faulty, if the defence names a condition under which it would hold. That is the skill being taught. Penalise nothing for finding fewer than twelve; the two texts contain more faults than most classes will exhaust.